

Vector Vortex Coronagraph Experiments in Vacuum Towards 10^{-10} Contrast

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ABSTRACT

Starlight suppression to levels of contrast of 1×10^{-10} with an internal coronagraph would allow for detection and spectral characterization of Earth-analogs. Many coronagraph architectures have been proposed to address this science case. Among them, the vector vortex coronagraph (VVC) stands out for its exquisite sensitivity especially at small angular separations. However, the VVC has yet to demonstrate 1×10^{-10} contrast in the laboratory. The limitation of VVCs with respect to the Lyot Coronagraph, which holds the contrast record, has been thought to be the vortex mask. Indeed, the mask fabrication imperfections limit how well the deformable mirrors can suppress starlight in the image during wavefront control. Furthermore, the polarization leakage inherent to the VVC has not been fully addressed as a source of incoherent light that limits this type of coronagraph's performance. Our new experiments in the Decadal Survey Testbed confirm these suspicions. Here we present the results of these experiments with a comprehensive characterization of our two best VVC masks.

Keywords: Exoplanets, High contrast imaging, Coronagraphs

1. INTRODUCTION

The search for habitable Earth analogs with the Habitable Worlds Observatory (HWO) necessitates a coronagraph design that efficiently provides extremely high contrast with sufficient throughput and robustness to small changes of the wavefront. The vortex coronagraph¹ is a promising solution that, in theory, provides remarkable raw contrast while retaining much of the throughput at small angular separations. Moreover, with a carefully chosen vortex topology the vortex coronagraph is able to deal with low-order wavefront aberrations. However, no vortex coronagraph has been demonstrated to yield 1×10^{-10} in laboratory conditions.

The manufacturing technology that has yielded the best results in terms of raw contrast for an optical vortex mask is with liquid crystal polymers (LCPs). The LCP is photo-aligned to produce a half-wave plate with a rotating optical axis about the center. A coronagraph with this kind of mask is called a vector vortex coronagraph (VVC). In this proceedings we present the status of the technology demonstration of the VVC at the High Contrast Imaging Testbeds (HCIT) with a particular focus on our current understanding of the limitations of these masks. This work is the continuation of the studies introduced in Ruane et al., 2022.² In Sec. 2 we introduce the main limitations of VVCs, in Sec. 3 we present the tools we use at HCIT to assess the performance of the masks, in particular, the Decadal Survey Testbed (DST), where the masks are put to the ultimate test of wavefront control. In Sec. 4 we show the results for the mask characterization of our two best masks and the DST results in narrowband and broadband light.

2. THE VECTOR VORTEX MASK LIMITATIONS

The laboratory performance of the VVC in terms of raw contrast is currently limited by the mask fabrication. Indeed, DST's performance at its best reaches a raw contrast of 4×10^{-10} in 10% BW light from 3 to 8 λ/D with a Classical Lyot Coronagraph (CLC).³ The VVC's best performance is 1.4×10^{-9} for the same bandwidth.² In Fig. 1, we show the typical limitations for the VVC. Let us take a closer look at these.

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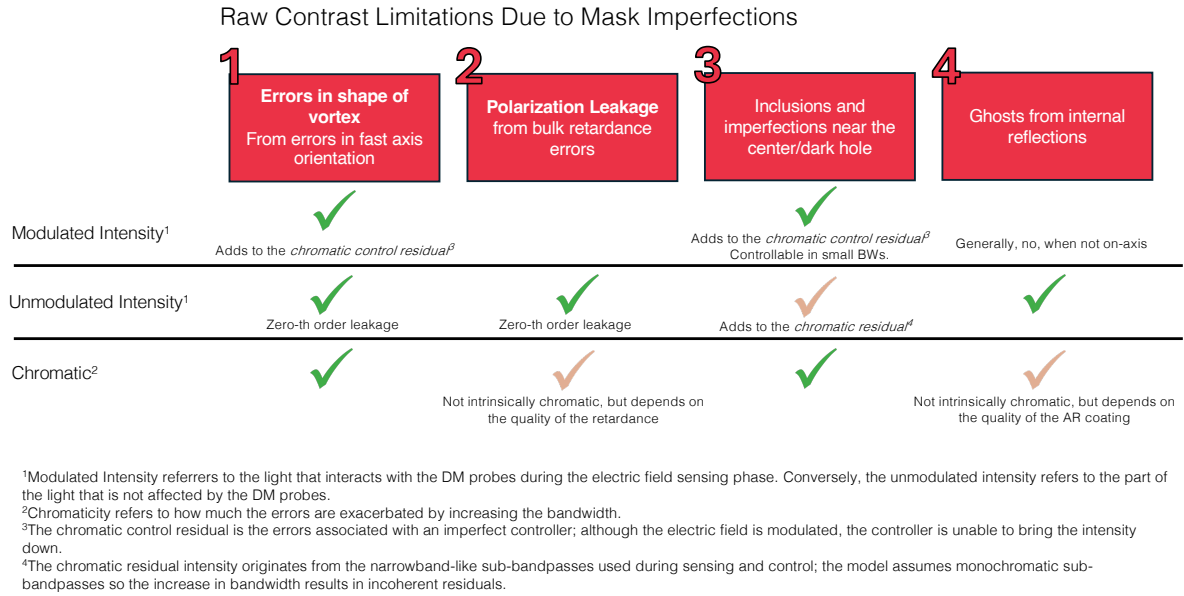


Figure 1. Summary of the main limitations to the raw contrast of a VVC originating from mask fabrication errors. The concepts of *chromatic residuals* and *chromatic control residuals* are taken from Seo et al., 2019.³

2.1 Vortex shape errors

The shape of the vortex in a VVC mask is achieved by rotating the optical axis of the half-wave plate about the center. This is achieved by photoaligning the LPC over a substrate. Of course, the ideal vortex shape is never attained, which introduces phase profile errors. First, the vortex shape inaccuracies introduce two types errors: (1) model errors, and (2) incoherent modes from deviations from the ideal vortex. The model errors will hinder the control, particularly when increasing the control bandwidth: this kind of limitation is known as the chromatic control residual.³

Seconds, deviations from the ideal vortex result in incoherent zero-th order leakage of similar nature to the polarization leakage, that is, the light appears in the image plane as a non-coronagraphic PSF. However, the polarization state of this light is identical to that of the off-axis source, so it is not possible to cancel it with the analyzer. In order to evaluate different kinds of errors in the vortex, and similarly to Ruane et al., 2019,⁴ and Desai et al., 2022,⁵ we perform a modal analysis of the azimuthal profile of the phase mask. In Fig. 2 we show the modal decomposition of three different kinds of errors. Note that this is only for azimuthal errors; an analytical analysis of both radial and azimuthal deviations from the ideal would be a rather difficult undertaking and beyond the scope of this discussion. The vortex profiles considered in Fig. 2 are:

- The ideal vortex (charge 6): the only mode present is the mode corresponding to the charge.
- Random Gaussian error of amplitude $\sigma=\pi/40$, with a frequency no higher than 20.
- Error in the vortex shape: a multiplying factor of amplitude $\sigma=\pi/40$ is added to the charge of the vortex.
- Sinusoidal error of amplitude $\sigma=\pi/40$ and frequency 5 added to the vortex profile.

The most unfavorable mode is mode 0. This is associated to the zero-th order leakage at the Lyot stop pupil plane, in other words, it ends up in the image plane as a non-coronagraphic PSF, and is not controllable by the deformable mirrors (DMs). Next, the uneven modes are not as pernicious, but they do contribute to leakage at the pupil plane that is not blocked by the Lyot stop. Finally, even modes are theoretically fully nulled at the Lyot stop plane, but modes under the nominal charge are more sensitive to low order aberrations.

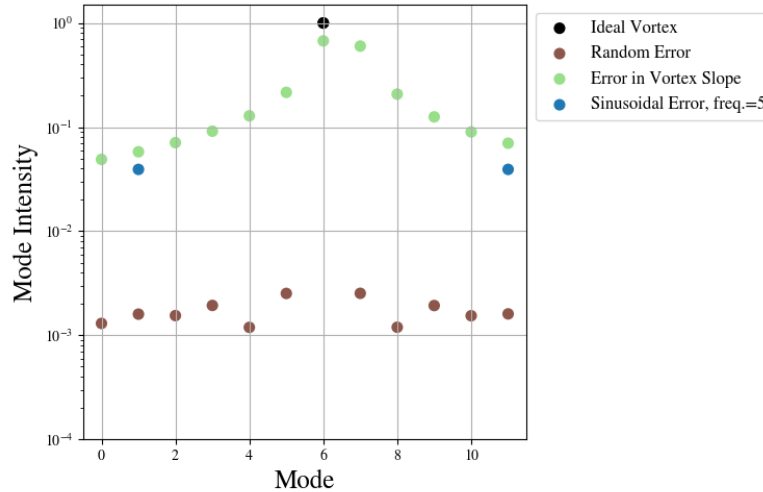


Figure 2. Modal decomposition of the vortex phase profile for the ideal case (*black*) and different cases of azimuthal errors. Ideally, the only mode present would be the mode associated to the charge of the vortex coronagraph. Errors in the azimuthal profile result in other modes emerging. The most pernicious mode is mode 0, followed by the uneven modes. Even modes smaller than the design vortex charge amplify the sensitivity to low order modes.

2.2 Polarization leakage

The polarization leakage originates from the imperfect retardance of the VVC half-wave plate. A perfect retardance of 180° is not attainable, and deviations from this number result in part of the light not *seeing* the vortex phase change. Therefore, this fraction of the light is let through as a flat wavefront through the Lyot stop and ends up in the image plane as non-coronagraphic PSF. This leakage is not controllable with the DMs and can only be mitigated by adding a circular analyzer downstream of the focal plane mask.

Ruane et al., 2022,² did a thorough analysis of the polarization leakage in VVC masks, we refer to that work for an in-depth examination of this phenomenon. In Ruane et al., 2022,² the intensity associated with the leakage is expressed as being proportional to the square of the error in the bulk retardance, $|c_L|^2 \sim (\epsilon_r/2)^2$. In practice, we find that a retardance of better than 1° is needed to reach better than 10^{-9} . This is assuming the circular analyzer can reach a polarization extinction of $\sim 1 \times 10^4$. Therefore, improving the optics in the circular analyzer is path to relaxing the requirements on the mask retardance.

The polarization leakage has received much attention due it being an intrinsic source of incoherent light for the VVC. However, it hasn't been shown yet to be the main source of incoherent light in broadband light. As we hope to show in Sec. 4, although the contrast floor in monochromatic and narrowband light is typically dominated by the polarization leakage, in broadband light other more chromatic sources of error have a comparable or larger contribution to the contrast floor.

2.3 Inclusions and imperfections near the center

Amplitude errors in the focal plane are a source of very chromatic speckle noise. In a VVC mask these typically originate from inclusions and the vortex center imperfections. The distinction of inclusions and imperfections is made by their nature: the former come from unwanted particles that deposit within the LCP during fabrication, and the latter from fabrication spin and coating imperfect photoalignment. Both, however, result in the same kind of error: amplitude error near or at the focal plane that hinder wavefront control. In broadband light these errors are not fully correctable and result in bright speckles that spill incoherent light into the dark hole.

2.4 Ghosts from internal reflections within the mask

A ghost image is an unwanted image of the source that lands on the image plane; if near enough to the dark hole, it can contribute to the incoherent light component. It is not correctable with wavefront control, and it is

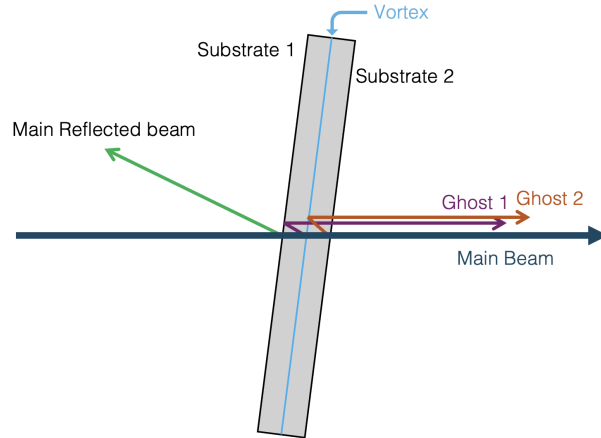


Figure 3. Schematic of the different reflection taking place at the VVC focal plane mask. Sizes and angles are not to scale. The focal plane mask is usually tilted with respect to the chief ray to avoid a ghost from the first reflection of the beam. The two ghosts originated from internal reflections have a key consequential distinction: ghost 1 (from the reflection in the upstream substrate) does not *see* the vortex, as opposed to ghost 2, which is a reflection of the beam after the vortex is imprinted on the beam. Therefore, ghost 1 is generally more detrimental.

hard to calibrate. Here we only consider ghosts caused by internal reflection within the VVC mask.

In Fig. 10 we show a schematic of the internal reflection that causes ghosts on the image plane. These come in two flavors: (1) front substrate reflection, and (2) back substrate reflection. A reflected beam from the back substrate has gone through the vortex center, and thus will form something close to a ring of fire at the Lyot stop pupil plane. The ghosts resulting from this kind of reflections are, thus, weaker. The ghosts from the reflection within the front substrate are mostly untouched by the vortex and will result in an off-axis-like PSF. These ghosts are more concerning.

3. LABORATORY SETUP

To assess the quality of our VVC masks with respect to the sources of errors presented in the previous section, we perform a series of tests in the HCIT facilities:

- High-resolution microscope imager: to look for inclusions and imperfections near the center.
- Muller Matrix Microscope Spectro-polarimeter: to measure the bulk retardance and the orientation of the optical axis.
- Spectrophotometer: to measure the AR coating of the mask substrates.
- Decadal Survey Testbed (DST): to measure the overall performance of the mask in terms of raw contrast.

3.1 High-resolution microscope imager

To look for inclusions and imperfections on the mask we use a Nikon ECLIPSE LV100 microscope. This microscope allows us to obtain a resolution of up to $0.12 \mu m$ per pixel. We take images in both reflected and transmitted light of the three mask planes: the two external substrate surfaces, and the vortex plane. In the vortex plane we look for inclusions that are product of contamination during the mask fabrication, and for the mask defect size. Indeed, at the center of the vortex, the LCPs have a harder time aligning to form the vortex, and a small defect forms at the center. Assessing the size of this defect is crucial; a mark as large as $0.2 \lambda/D$ will severely limit the achievable raw contrast.

Future work will entail adding measurements on the microscope to the model, i.e. having a model of the three-plane mask with the imaged errors, in order to predict performance.

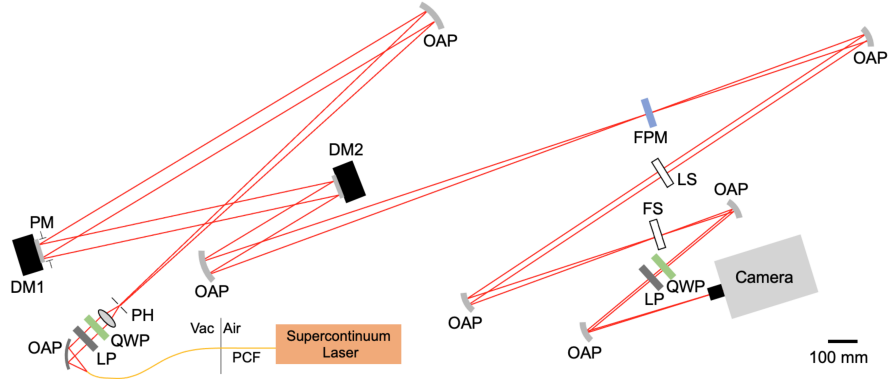


Figure 4. Optical layout of the Decadal Survey Testbed (DST) at HCIT. Taken from Ruane et al. 2022.²

3.2 The Muller matrix microscope spectro-polarimeter

The Muller matrix spectro-polarimeter (MMSP) gives us the ability to measure the Muller matrix at each resolution element near the vortex center. We can thus get a measurement of the bulk retardance and the orientation of the fast axis near the vortex center. Our MMSP is an Axometrics AxoStep; we typically work with a resolution of $5.8 \mu\text{m}$ per pixel, and over a spectral bandwidth of 400 to 800 nm with a spectral resolution of 5 nm .

3.3 The spectrophotometer

To characterize the reflectivity of the front and back surfaces of our vortex masks we have at our disposal a Perkin Elmer Lambda 1050 UV/VIS/NIR Spectrophotometer in the HCIT high-bay. However, this spectrophotometer has not been yet used to characterize our vortex masks. Future work will involve using it for this purpose. The current setup to characterize the reflectivity is in the DST chamber, using the integrated intensity reading from the wavefront sensor camera (reflected beam from the mask) and the reading from the science camera (transmitted beam). This is an estimation of the reflectance, which includes uncertainties due to, firstly, the differential QE of both cameras, plus the losses associated with the non-common path. However, with this measurement we look to assess the spectral profile of the reflectivity curve in order to detect peaks and valleys in reflectivity that would point to chromatic behaviour of the incoherent ghost.

3.4 DST

The wavefront control experiments in the DST are the ultimate test of a VVC mask since the raw contrast is the main metric for this technology demonstration. In Fig. 4 we show the layout of DST for the experiments presented in this work. Details of DST can be found in Patterson et al., 2019.⁶ Relevant to this work are the DMs: two Northrop Grumman's AOA Xinetics (AOX) DMs with 48×48 actuators, the focal ratio at the focal plane where the vortex mask sits is 32.7, the Lyot stop blocks 80% of the beam diameter. A circular polarizer and analyzer set is placed at the source assembly and at the end of the optical path respectively to allow for one polarization state only. The linear polarizers are Thorlabs LPVIS-100, and the quarter-wave plates are from Tower Optical (achromatic mounted 25.4 mm). DST is operated in vacuum to achieve maximum stability; a working pressure of ~ 0.1 mTorr is achieved and maintained for weeks. The temperature variations are in the order of ~ 10 mK in a cadence of 24h.

The source that simulated the star is a supercontinuum laser source (NKT SuperK EXTREME) followed by a tunable filter (NKT SuperK Extreme VARIA). The tunable filter allows for a bandwidth of $\sim 1\%$. The camera is an Oxford Instruments Andor Neo 5.5.

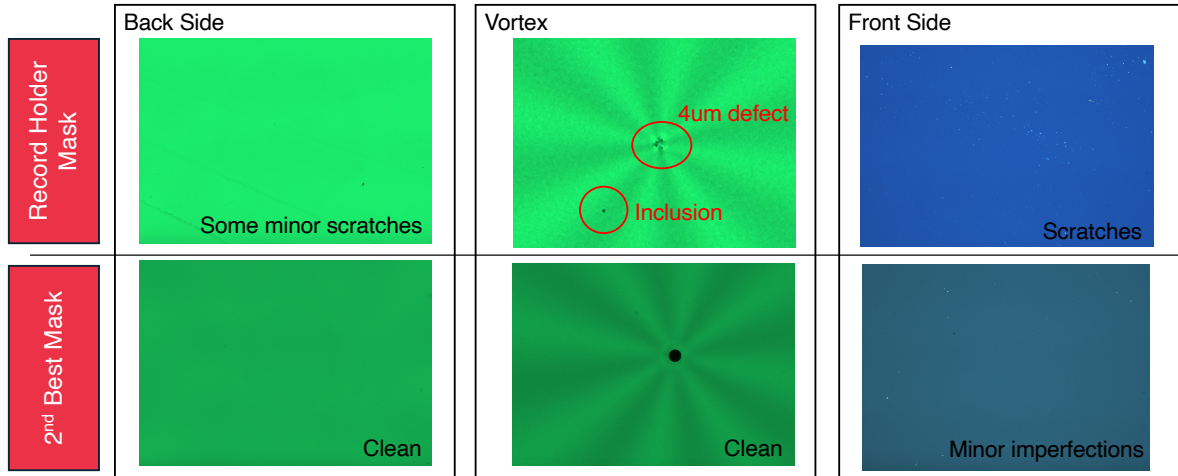


Figure 5. Images from HCIT's Nikon ECLIPSE LV100 microscope for the two best VVC masks. The *Second Best Mask* is significantly cleaner in all three surfaces with respect to the *Record Holder Mask*. In particular, in the vortex plane, the central defect and an inclusion are clearly detectable; however, even given these errors, the *Record Holder Mask* performs better in the testbed in terms of raw contrast.

4. RESULTS

In this section we will present the mask characterization and DST performance results for our two best masks. The part numbers used for these masks internally are rather long, so, for clarity purposes, we'll call them: (1) *Record Holder Mask*, and (2) *Second Best Mask*. As we'll show in this section, these two masks are limited by different phenomena, but reach comparable performance in broadband light at DST. This makes for a insightful case study to better understand their strengths and limitations, and project these lessons to future manufacturing.

4.1 Inclusions and imperfections

The high-resolution microscope allows us to look for visible imperfections in reflection and transimition in the three planes of the VVC mask. In Fig. 5 we show some images from the microscopes for the two masks. These masks, as expected, are relatively devoid of major errors, however, some imperfections were found at high resolution. In particular, for the *Record Holder Mask*. This mask exhibits a central defect of $\sim 4 \mu m$ and an inclusion at a separation close to the working angular separations for our dark holes. The size of the central defect with respect to the beam size is concerning at first glance, however, given the performance of this mask we find that, although it may be a limitation in broadband light, it is not the main limitation in narrowband and monochromatic, as it will be shown in the following sections.

4.2 Bulk retardance

The AxoStep MMSP provides us with the Muller matrix measurements for each resolution element of the MMSP. With these measurements we can compute the bulk retardance for each resolution element. For an ideal VVC mask, the bulk retardance would be that of an ideal half-wave plate, i.e. 180° retardance. Any deviations from this will result in polarization leakage. In Fig. 6 we show the retardance of our two best masks near the vortex center at 650 nm , and in Fig. 7 we show the bulk retardance as a function of wavelength. The *Record Holder Mask* has a significantly better bulk retardance from 600 to 650 nm , which are the design wavelengths. Furthermore, it reaches $< 1^\circ$ retardance error in a relatively large bandwidth. Therefore we expect the *Record Holder Mask* to have a significantly smaller contribution of the polarization leakage.

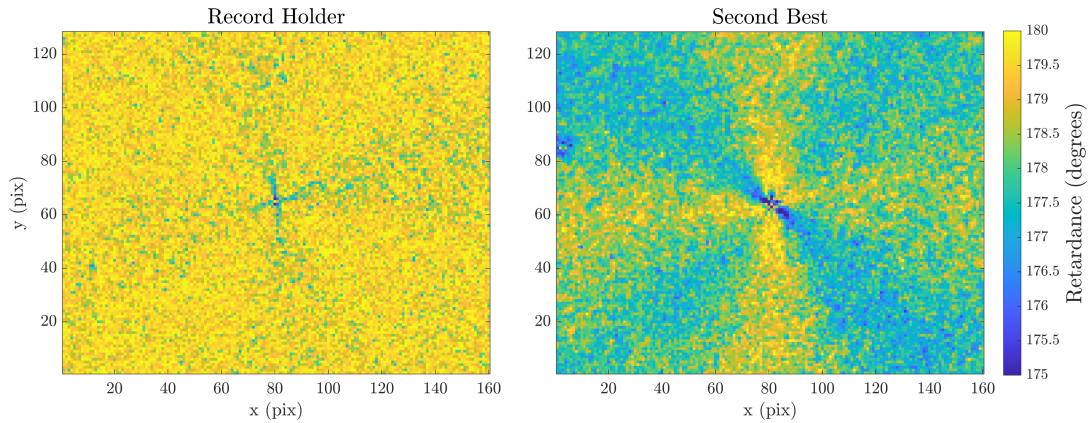


Figure 6. Retardance of our two best VVC masks, as measured by the AxoStep Muller Matrix Microscope at 630 nm. The closer the retardance is to half-wave, i.e. 180° , the smaller the polarization leakage. Therefore, the *Record Holder Mask*, which has a significantly better retardance with respect to the *Second Best Mask*, will have a smaller polarization leakage.

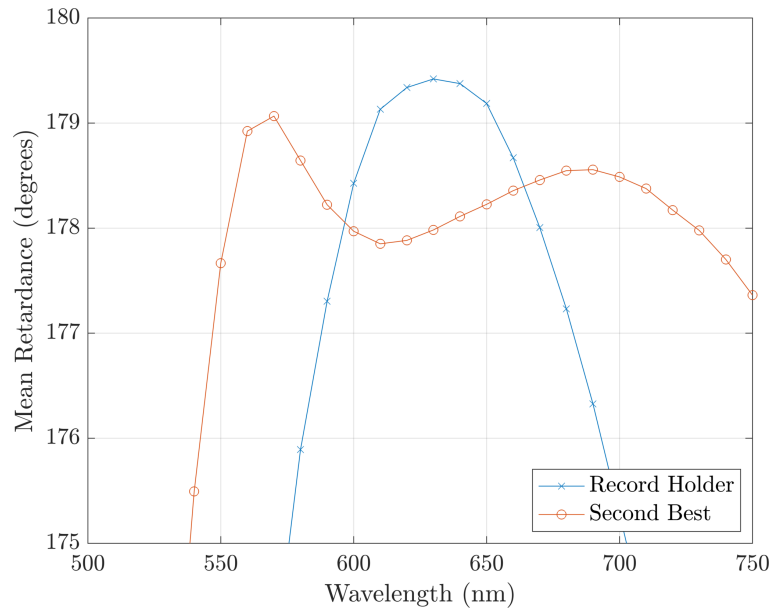


Figure 7. Bulk retardance as a function of wavelength for our two best VVC masks, as measured by the AxoStep Muller Matrix Microscope. The *Record Holder Mask* achieves a very good retardance error, $< 1^\circ$, over a relatively large bandwidth.

4.3 Vortex profile

In a similar way to the bulk retardance, the vortex profile, or the orientations of the fast axis, can be computed from the Muller matrix measurement from the AxoStep MMSP. In Fig. 8 we show the measured vortex profile compared to the ideal vortex; the plot that displays the computed difference has two contours: a first contour of $\pi/40$ and a second one of $\pi/20$.

Considering the discussion in Sec. 2.1, and, in particular, Fig. 2, we can roughly assess the upper limit of the effect of these errors in vortex shape. In Fig. 2 we see that a uniform vortex shape error over the whole mask is the worst case scenario and that random errors are not as detrimental. In Fig. 8 we can see that the errors are

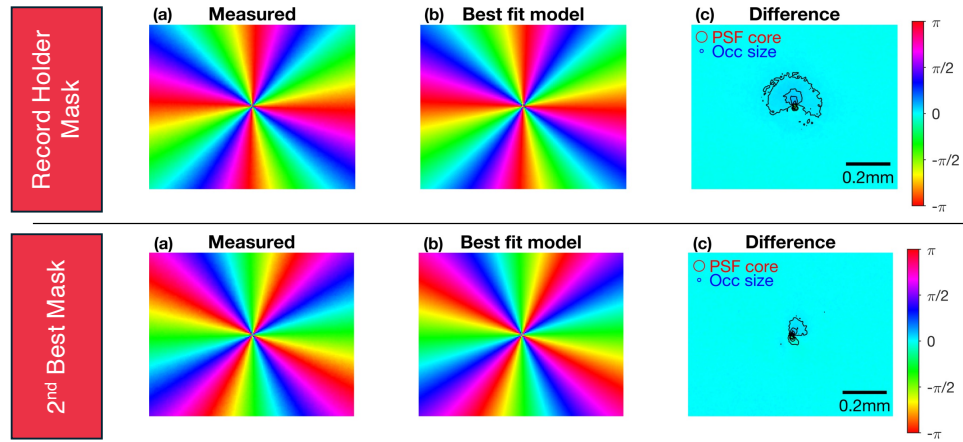


Figure 8. Comparison of the measured vortex profile, or orientation of the fast axis around the vortex center, for our two best masks. The first column is the measured profile computed from the AxoStep Muller Matrix Microscope. The second column is the ideal vortex model fit to the measurement, and the third column is the difference between the measurement and the model. The two contours are $\pi/40$ and $\pi/20$. The errors are small for both, the *Second Best Mask* having a closer to ideal shape overall.

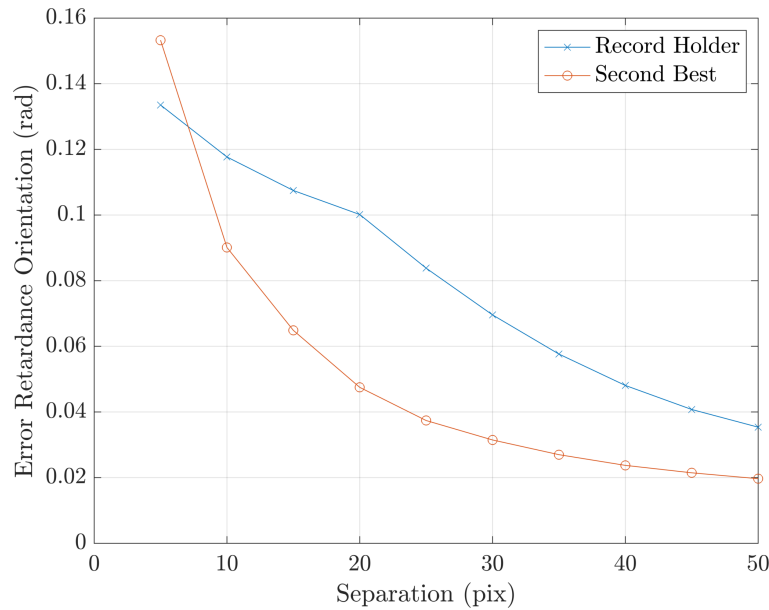


Figure 9. Deviation of the measured vortex shape with respect to the ideal vortex as a function of separation for our two best VVC masks. The *Second Best Mask* has an overall more ideal phase profile as measured by the AxoStep Muller Matrix Microscope.

somewhere between random and uniform, and only above $\pi/40$ near the center. This last consideration poses a source of uncertainty in this analysis that would be hard to disentangle; we can intuitively speculate that, although near the center of the vortex errors have a larger effect, given the fact that these are not as bad at larger separations, we can expect that roughly no more than 1% of the modal power falls in modes other than the vortex charge mode. Again, this is a rough upper limit based on an idealized set of errors from Fig. 2. Ideally, a full model based on the measurements presented here would help us get a reliable estimate of the effects of these errors in broadband light.

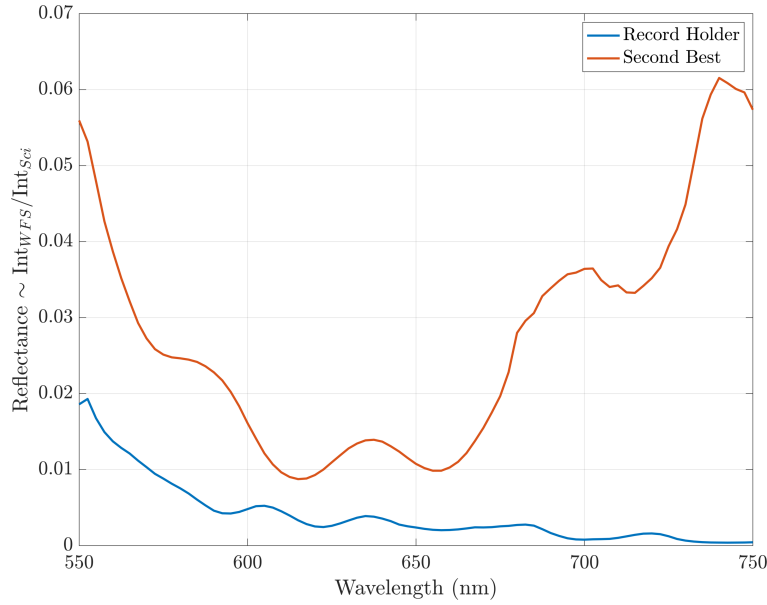


Figure 10. Estimation of the reflectance for the two best masks, computed as the intensity ratio between the pupil imaged at the wavefront sensing camera (see Fig. 4) and the pupil imaged in the science camera.

4.4 Ghosts

As explained in Sec. 3.3 the spectrophotometer has not been available for our use so we are unable to show results for our two masks under this device. However, as explained in the same section, we compute a proxy of the reflectance with the intensity ratio of the pupil imaged in the wavefront sensing camera and the science camera. In Fig. 10 we show these measurements for both masks, as a function of wavelength. A noteworthy caveat with respect to the *Second Best Mask*: the reflection off the mask exhibits multiple beams, some of which are clipped by the wavefront sensing camera imaging system, hence we had to correct the intensity from this measurements. We add an *ad hoc* factor of 2 to compensate. Again, with this test we're not trying to get precise photometry, we hope to do so with the spectrophotometer, but to get an idea of the reflectance profile.

The quality of the coating of the *Record Holder Mask* is significantly better with respect to the *Second Best Mask*. An assessment of the ghosts should include the wedge of the substrate, which we currently don't have any data for. That being said, we generally expect to have a worst ghost situation with the *Second Best Mask*.

4.5 Raw contrast in DST

The raw contrast achieved in our vacuum chamber testbed, DST, is the most relevant metric for this technology demonstration. The target average raw contrast is 1×10^{-10} in 10% bandwidth. To dig a dark hole we use the wavefront control algorithm electric field conjugation, or EFC,⁷ the gold standard for wavefront control in high contrast imaging. We run EFC in all the HCIT testbeds with the Fast Linear Least-Squares Coronagraph Optimization (FALCO)⁸ software package*.

4.5.1 Monochromatic result

Before diving into the narrowband and broadband analysis of the two masks, let us present the monochromatic EFC results for the *Record Holder Mask*. To perform these experiments we switched the white-light source and tunable filter for a laser source (Thorlabs LP637-SF70 diode laser). We consistently reach levels of raw contrast below 3×10^{-10} . To validate this result we ran EFC four times from a flat DM configuration and reached an average raw contrast of $2.44 \pm 0.16 \times 10^{-10}$. This represents the best result ever for a vortex coronagraph in monochromatic light.

*<https://github.com/ajeldorado/falco-matlab>

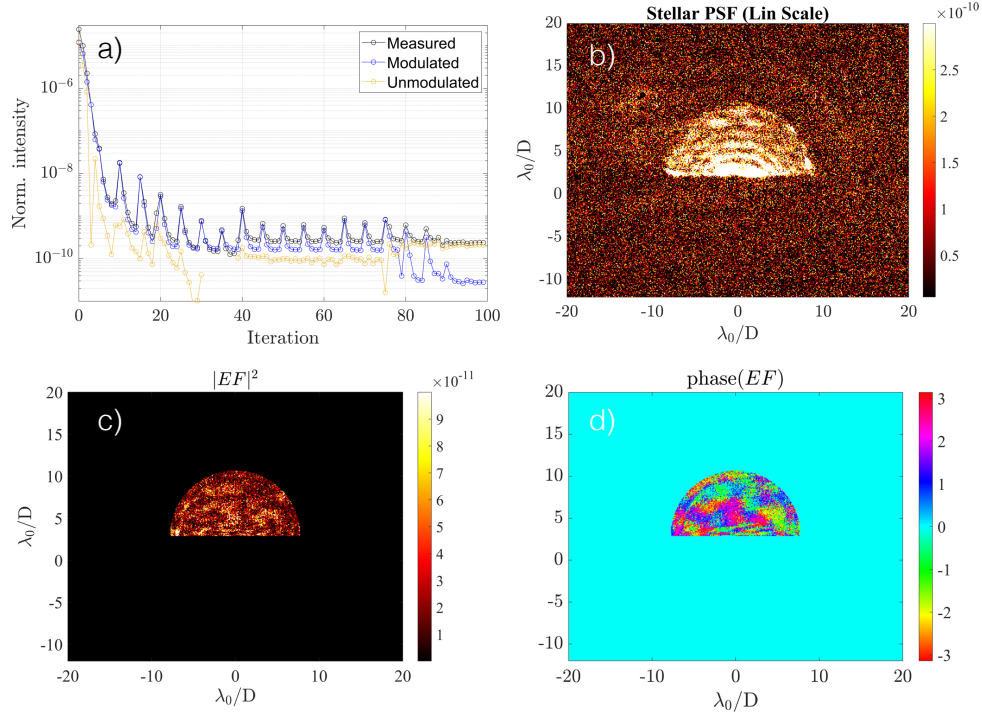


Figure 11. Monochromatic EFC results for the *Record Holder Mask* in one of the runs that achieved $\sim 2.5 \times 10^{-10}$. *a)* Normalized intensity for the full EFC run; the regularly spaced increases in intensity are due to the β -bumping strategy to reduce the contrast floor.⁹ *b)* Image of the dark hole after the last EFC iteration; the concentric uncorrected rings indicate that the limitation is due to the polarization leakage. *c)* and *d)* Intensity and phase computed from the estimation of the electric field after the last EFC iteration; the structure of the speckles may indicate that the limit in the modulated component is dominated by the DM electronics noise.¹⁰

In Fig. 11 we show the EFC result from one of the four runs. The evenly spaced positive jumps in contrast seen in Fig. 11a are due to the regularization scheme that is typically used during an EFC run; this is also known as β -bumping.⁹ As it can be noted from Fig. 11b, concentric rings are limiting the contrast achievable. These are due to the polarization leakage. The modulated component is shown in Fig. 11c and d as the reconstructed electric field estimated with pair-wise probing, EF_s after the last iteration of EFC. We plot the associated intensity of the sensed electric field, $|EF_s|^2$, and its phase. Since at the last iteration of EFC the final raw contrast is dominated by the incoherent polarization leakage, the intensity of $|EF_s|^2$ is very low, $\sim 3 \times 10^{-11}$. The speckle-y structure of this EF_s , combined with the fact that the four EFC runs exhibit a seemingly random distribution from run to run, may indicate that the modulated limit is the DM electronics noise. This is a noise associated with the least-significant-bit of the DM electronics, an intrinsic source of noise when using DMs.¹⁰

4.5.2 Narrowband results

With DST's nominal setup, this is with the white-light source and tunable filter connected to the source, we can perform narrowband light, $\sim 1\%$ BW, EFC runs for different wavelengths. By sweeping a set of wavelengths around the design wavelength we can assess the quality of the mask and predict its limitations in broadband light. In narrowband, we expect to be limited by sources of incoherent light that are not controllable, e.g. the polarization leakage and ghosts. We find this to be true for our two best masks: the *Record Holder Mask* is limited by the polarization leakage, as we saw in the previous section, and the *Second Best Mask* is limited by a ghost from an internal reflection within the VVC mask. In Fig. 12 we show the EFC results for central

Narrowband, $\sim 1\%$ at 625 nm

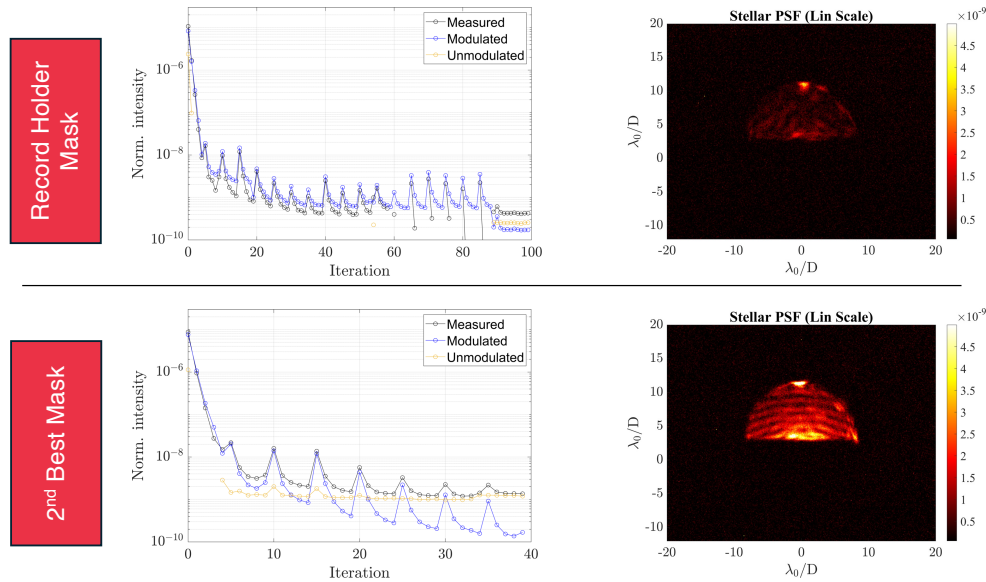


Figure 12. EFC results for the narrowband, $\sim 1\%$ bandwidth, at 625 nm for the *Record Holder Mask* (top) and the *Second Best Mask* (bottom). The left column shows the normalized intensity as a function of iterations, with a breakdown of the total intensity in its modulated and unmodulated components. The difference between these two components gives us a hint into what is the nature of the contrast limitation. The *Record Holder Mask* is limited by the polarization leakage and some chromatic control residual due to the deviation from monochromatic. The *Second Best Mask* is limited by an off-axis ghost at $\sim 12 \lambda/D$ South from the center.

wavelength $\lambda_0 = 625$ nm. The *Record Holder Mask* does not exhibit the strong concentric rings that appeared in monochromatic light; this is probably due to chromatic control residual since the modulated component has a comparable intensity to the unmodulated component. This component stems from increasing the bandwidth. The *Second Best Mask* is limited by an off-axis ghost as indicated by the incoherent rings centered at $\sim 12 \lambda/D$ below the center. This contrast floor is at the level of 10^{-9} .

In Fig. 13 we show the final normalized intensity after an EFC run as a function of wavelength. The performance of both masks is rather flat around the design wavelengths, from 620 to 660 nm. This would indicate that, if the limitations in narrowband are the same as in broadband, we should see similar contrast floors. However, this is not the case as shown in the next section.

4.5.3 Broadband results

We repeat the experiments described in the section above for the broadband case, with 10% bandwidth light. Fig. 14 shows the EFC results, and Fig. 15 the EFC sweep of wavelengths.

An important result is that the *Record Holder Mask* performs significantly worse in broadband: it reaches 3×10^{-10} in monochromatic light, and only $\sim 1.5 \times 10^{-9}$ in broadband light. This is attributed to sources of incoherent light that are highly chromatic, e.g. errors in the vortex shape and inclusions and imperfections in the mask. Indeed, in the left column of Fig. 14 we can assess the weight of the modulated versus unmodulated components of the light to give us an idea of the nature of the contrast floor. For the *Record Holder Mask*, the contribution of the modulated component is significant, $\sim 50\%$, which indicates that the chromatic control residual stemming from errors in the vortex shape and from amplitude errors in, or near, the vortex plane, may have a large effect in the contrast floor. Distinguishing between the two is not trivial, and modelling will be needed.

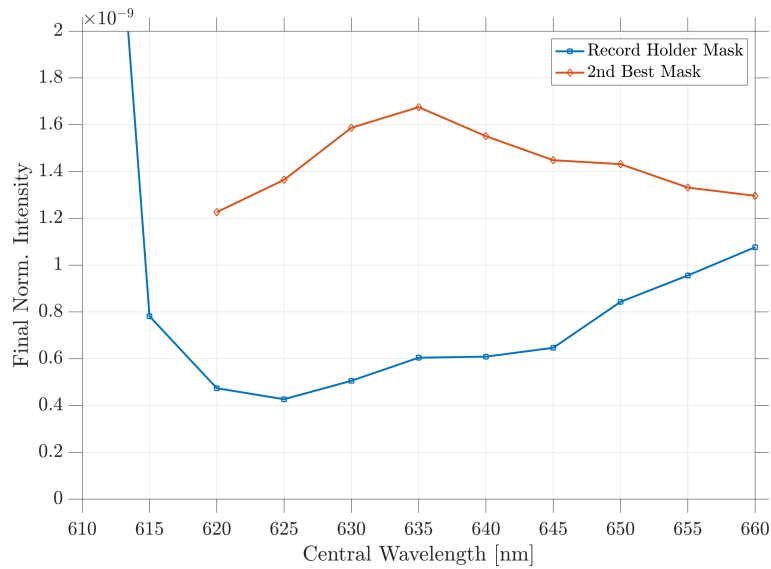


Figure 13. Final normalized intensity of different EFC runs for a set of central wavelengths in narrowband light. The contrast limitations for the two masks are of different nature: the polarization leakage for the *Record Holder Mask*, and a ghost for the *Second Best Mask*. These contrast floors also differ a factor of ~ 4 in intensity. Given the uniformity of these curves, if the same phenomena are limiting the contrast in the broadband case, we would expect to reach similar limits of contrast.

Broadband, 10% at 625 nm

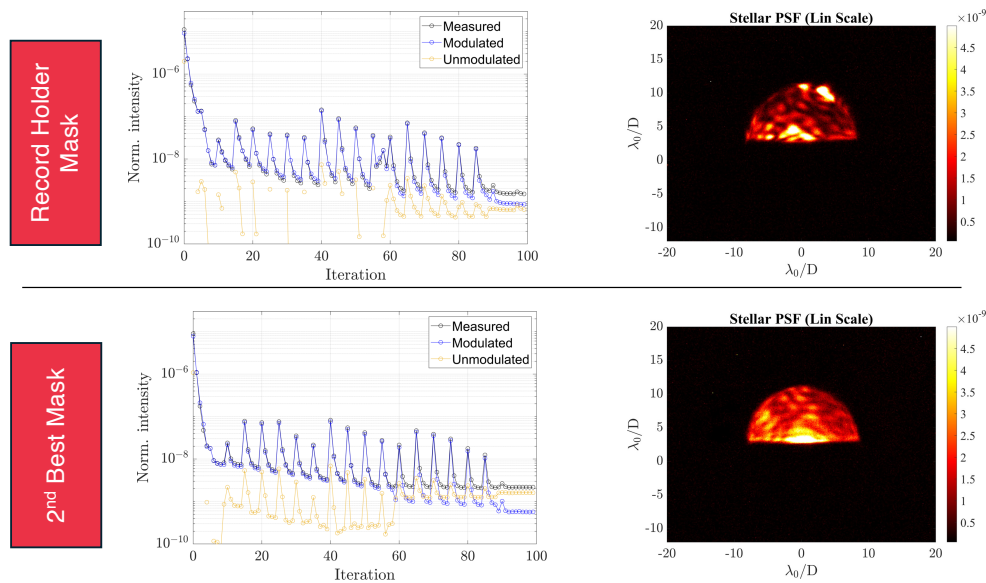


Figure 14. Same as Fig. 12 but for the broadband case, 10% bandwidth at 625 nm. An interesting difference between the two masks is the contribution of the modulated component of the light to the total intensity in the final iterations (left column); the *Record Holder Mask* has a relatively large contribution of the modulated component, $\sim 50\%$, whereas the *Second Best Mask* these contributions are very small, $< 10\%$.

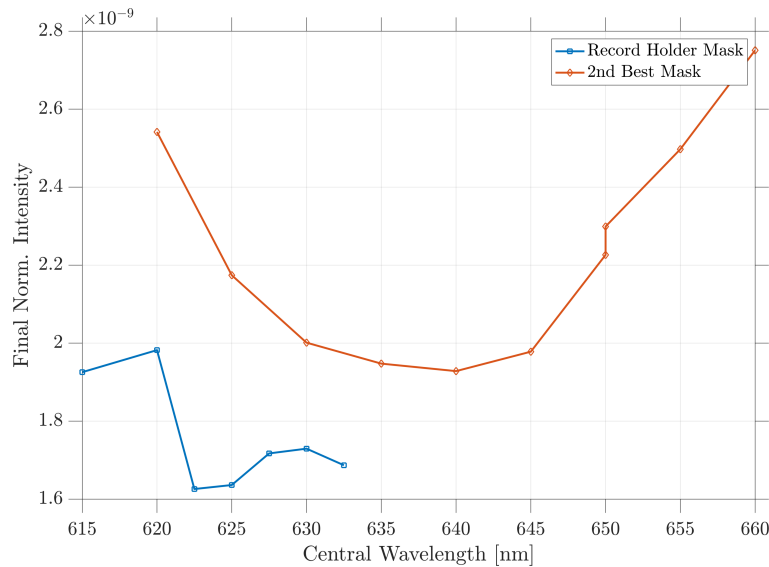


Figure 15. Same as Fig. 13 but for the broadband case, 10% bandwidth at 625 nm. In this case, though, both masks perform to similar levels of contrast. This is due to the *Second Best Mask* being less affected by chromatic control residuals since (1) its vortex shape is closer to ideal and (2) it is cleaner at the vortex plane within the mask.

As for the *Second Best Mask*, the modulated component in the last iterations (left column of Fig. 14) is very small, $< 10\%$. Furthermore, the different in contrast levels between the narrowband results and the broadband results is relatively small. This would indicate that the limitations are similar in both cases, and the contribution of the chromatic control residual is relatively small.

These results are consistent with the mask characterization described in previous sections. Indeed, the *Second Best Mask* has a more perfect vortex shape and is very clean in terms of inclusions and imperfections as seen through the microscope. On the other hand, the *Record Holder Mask* is superior in terms of the polarization leakage. This translates to the *Record Holder Mask* performing much better in narrowband light, but both masks performing similarly in broadband since the *Second Best Mask* has less contributions of the chromatic control residual to the contrast floor.

5. CONCLUSIONS

In this work we have presented the latest laboratory results of the vector vortex coronagraph (VVC) at HCIT, and the mask characterization procedures for the VVC focal plane masks. We have presented the new monochromatic raw contrast record for a VVC, 2.44×10^{-10} . We have compared the two best performing VVC masks to assess different phenomena that typically limit the performance of the VVC. The *Record Holder Mask* has an exquisite retardance that allows it to reach raw contrast levels near 10^{-10} in monochromatic and narrowband light. However, in broadband light this mask is limited by both modulated and unmodulated light. This hints at the fact that chromatic control residuals from deviations from the ideal vortex phase profile and inclusions and imperfections near the vortex center, may be increasing the contrast floor when increasing the bandwidth. On the other hand, the *Second Best Mask*, performs to comparable levels of raw contrast in both narrowband and broadband light. This is thanks to a close-to-ideal vortex phase profile and the cleanliness of the mask near the center of the vortex. However, the retardance of this mask and a nearby ghost limit its performance to worse than 10^{-9} raw contrast.

These results help us understand the VVC limitations better, and guide the manufacturing process that will bring improvements in mask quality. We are working on improving our characterization capabilities with new equipment, processes and modeling efforts given the importance of fully understanding the limitations of our masks, not just for the VVC concept but for any coronagraph design. Indeed, obtaining a coronagraph mask

that brings us down to the very much sought 1×10^{-10} raw contrast in broadband light is a critical path for a viable Habitable Worlds mission.

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REFERENCES

- [1] Mawet, D., Riaud, P., Absil, O., and Surdej, J., “Annular Groove Phase Mask Coronagraph,” *Astrophys. J.* **633**, 1191–1200 (2005).
- [2] Ruane, G., Riggs, A. J. E., Serabyn, E., Baxter, W., Mejia Prada, C., Mawet, D., Noyes, M., Poon, P. K., and Tabiryan, N., “Broadband vector vortex coronagraph testing at NASA’s high contrast imaging testbed facility,” in [*Space Telescopes and Instrumentation 2022: Optical, Infrared, and Millimeter Wave*], Coyle, L. E., Matsuura, S., and Perrin, M. D., eds., *Society of Photo-Optical Instrumentation Engineers (SPIE) Conference Series* **12180**, 1218024 (Aug. 2022).
- [3] Seo, B.-J., Patterson, K., Balasubramanian, K., Crill, B., Chui, T., Echeverri, D., Kern, B., Marx, D., Moody, D., Prada, C. M., Ruane, G., Shi, F., Shaw, J., Siegler, N., Tang, H., Trauger, J., Wilson, D., and Zimmer, R., “Testbed demonstration of high-contrast coronagraph imaging in search for Earth-like exoplanets,” in [*Techniques and Instrumentation for Detection of Exoplanets IX*], Shaklan, S. B., ed., **11117**, 599 – 609, International Society for Optics and Photonics, SPIE (2019).
- [4] Ruane, G., Mawet, D., Riggs, A. J. E., and Serabyn, E., “Scalar vortex coronagraph mask design and predicted performance,” in [*Society of Photo-Optical Instrumentation Engineers (SPIE) Conference Series*], *Society of Photo-Optical Instrumentation Engineers (SPIE) Conference Series* **11117**, 111171F (Sept. 2019).
- [5] Desai, N., Llop-Sayson, J., Bertrou-Cantou, A., Ruane, G., Eldorado Riggs, A. J., Serabyn, E., and Mawet, D., “Topological designs for scalar vortex coronagraphs,” in [*Space Telescopes and Instrumentation 2022: Optical, Infrared, and Millimeter Wave*], Coyle, L. E., Matsuura, S., and Perrin, M. D., eds., *Society of Photo-Optical Instrumentation Engineers (SPIE) Conference Series* **12180**, 121805H (Aug. 2022).
- [6] Patterson, K. D., Seo, B.-J., Balasubramanian, K., Chui, T., Crill, B., Gill, J., Kern, B., Lam, R., Marx, D., Moody, D., Prada, C. M., Ruane, G., Rupp, J., Shaw, J., Shi, F., Siegler, N., Tang, H., Trauger, J., White, V., Wilson, D., Yee, K., and Zimmer, R., “Design description and commissioning performance of a stable coronagraph technology development testbed for direct imaging of Earth-like exoplanets,” in [*Techniques and Instrumentation for Detection of Exoplanets IX*], Shaklan, S. B., ed., **11117**, 111171U, International Society for Optics and Photonics, SPIE (2019).
- [7] Give’on, A., Kern, B., Shaklan, S., Moody, D. C., and Pueyo, L., “Broadband wavefront correction algorithm for high-contrast imaging systems,” in [*Astronomical Adaptive Optics Systems and Applications III*], Tyson, R. K. and Lloyd-Hart, M., eds., **6691**, 63 – 73, International Society for Optics and Photonics, SPIE (2007).
- [8] Riggs, A. J. E., Ruane, G., Sidick, E., Coker, C., Kern, B. D., and Shaklan, S. B., “Fast linearized coronagraph optimizer (FALCO) I: a software toolbox for rapid coronagraphic design and wavefront correction,” in [*Space Telescopes and Instrumentation 2018: Optical, Infrared, and Millimeter Wave*], *Society of Photo-Optical Instrumentation Engineers (SPIE) Conference Series* **10698**, 106982V (Aug. 2018).
- [9] Sidick, E., Seo, B.-J., Kern, B., Marx, D., Poberezhskiy, I., and Nemati, B., “Optimizing the regularization in broadband wavefront control algorithm for WFIRST coronagraph,” in [*Techniques and Instrumentation for Detection of Exoplanets VIII*], Shaklan, S., ed., **10400**, 587 – 599, International Society for Optics and Photonics, SPIE (2017).
- [10] Ruane, G., Echeverri, D., Bendek, E., Kern, B. D., Marx, D., Mawet, D., Prada, C. M., Riggs, A. J. E., Seo, B.-J., Serabyn, E., and Shaklan, S., “Microelectromechanical deformable mirror development for high-contrast imaging, part 2: the impact of quantization errors on coronagraph image contrast,” *Journal of Astronomical Telescopes, Instruments, and Systems* **6**, 045002 (Oct. 2020).